

Loan Prediction Model — Analysis Report

A comprehensive evaluation of machine learning approaches for automated loan approval prediction



Project Overview

This project develops a binary classification system to predict loan approval outcomes using machine learning. Two distinct algorithms—Logistic Regression and Random Forest Classifier—were trained and rigorously compared using identical preprocessing pipelines and evaluation metrics.

The target variable, **loan_status**, represents a binary outcome where 1 indicates loan approval and 0 indicates rejection. This classification task enables financial institutions to automate risk assessment while maintaining accuracy and minimizing both false approvals and missed opportunities.

Binary Classification

Approved (1) vs Rejected (0)

Primary Objective

Accurate risk prediction with minimal false positives



Dataset Summary

The dataset comprises 5,000 loan applications with 12 predictive features, split into 80% training (4,000 records) and 20% testing (1,000 records) with a random state of 42 for reproducibility.

5K

Total Records

Loan applications analyzed

12

Features

Predictive variables used

71.4%

Approved

Class 1 instances

28.6%

Rejected

Class 0 instances

Feature Categories

Numerical Features

- person_age (applicant age)
- person_income (annual income)
- person_emp_length (years employed)
- loan_amnt (requested amount)
- loan_int_rate (interest rate %)
- loan_percent_income (% of income)
- cb_person_cred_hist_length (credit history)

Categorical Features

- person_gender (encoded)
- person_home_ownership (encoded)
- loan_intent (purpose, encoded)
- person_education (highest level, encoded)
- previous_loan_defaults_on_file (prior defaults, encoded)



Moderate Class Imbalance: With 71% approved loans, a naive classifier predicting "Approved" always would achieve 71% accuracy. Therefore, accuracy must be evaluated alongside precision, recall, and AUC-ROC metrics.

All categorical columns were encoded using scikit-learn's LabelEncoder before model training to ensure compatibility with both algorithms.

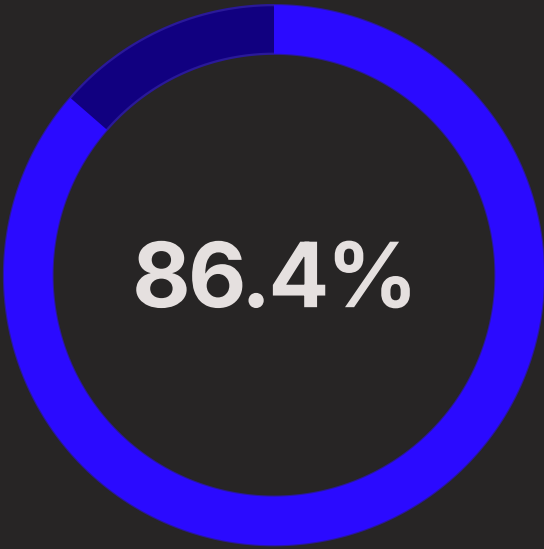
Model 1 — Logistic Regression

Algorithm
Logistic Regression with lbfgs solver

Iterations
Maximum 1,000 iterations allowed

Convergence
Reached solution within iteration limit

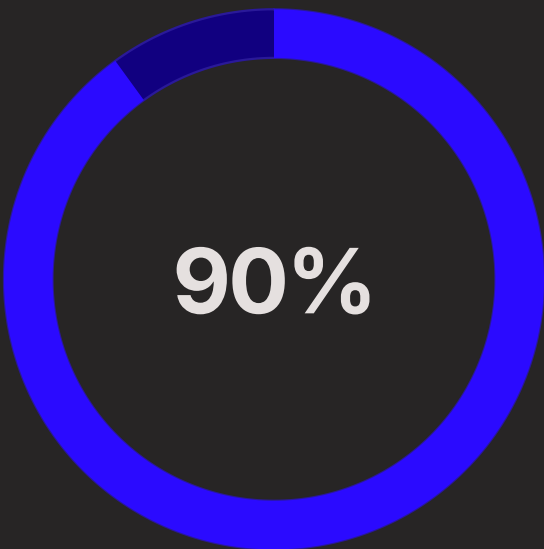
Performance Metrics



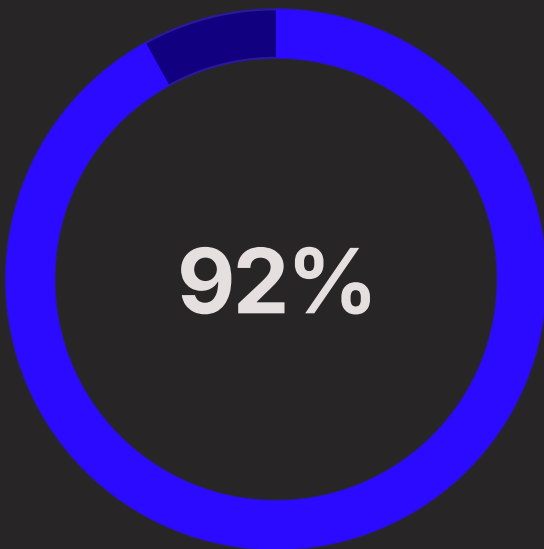
Accuracy
Overall correctness



AUC-ROC
Discrimination ability



Precision (Approved)
Quality of approvals



Recall (Approved)
Capture of valid loans

Classification Report

Class	Precision	Recall	F1-Score	Support
Rejected (0)	0.76	0.72	0.74	265
Approved (1)	0.90	0.92	0.91	735
Weighted Avg	0.86	0.86	0.86	1,000

Confusion Matrix

	Predicted Rejected	Predicted Approved
Actual Rejected	190	75
Actual Approved	61	674

Interpretation

Correctly identified 190 rejected loans and 674 approved loans. Generated 75 false positives (risky approvals) and 61 false negatives (missed opportunities).

Model 2 — Random Forest Classifier



Ensemble Method
100 decision trees voting

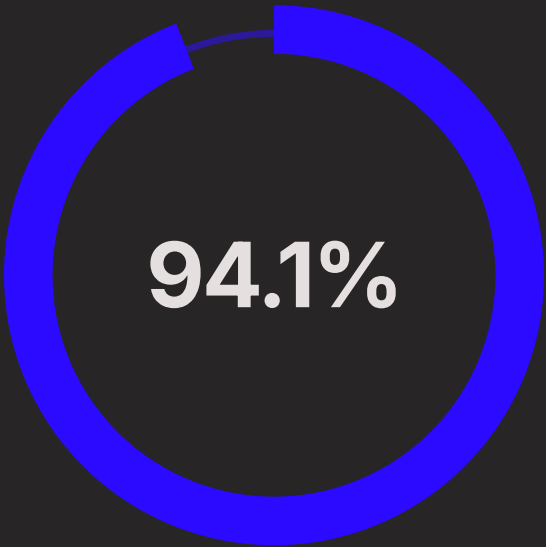


Random State
42 for reproducibility



Objective
Reduce variance through aggregation

Performance Metrics



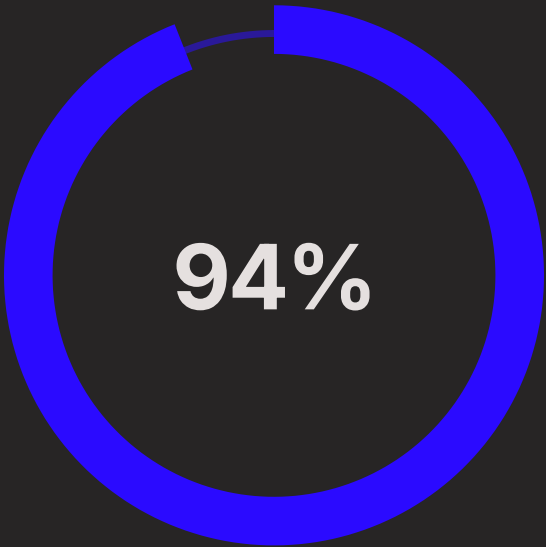
Accuracy

7.7 points above Logistic Regression



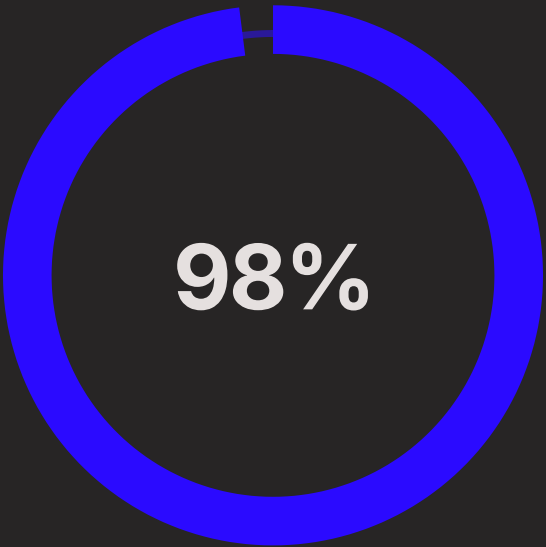
AUC-ROC

Near-perfect discrimination



Precision (Approved)

High-quality approvals



Recall (Approved)

Nearly all valid loans captured

Classification Report

Class	Precision	Recall	F1-Score	Support
Rejected (0)	0.93	0.84	0.88	265
Approved (1)	0.94	0.98	0.96	735
Weighted Avg	0.94	0.94	0.94	1,000

Confusion Matrix

	Predicted Rejected	Predicted Approved
Actual Rejected	222 	43 
Actual Approved	16 	719 

Interpretation

Correctly identified 222 rejected loans and 719 approved loans. Generated only 43 false positives and 16 false negatives—significantly better than Logistic Regression.

Model Comparison

Random Forest consistently outperforms Logistic Regression across all evaluation metrics, demonstrating superior predictive capability for loan approval classification.

Metric	Logistic Regression	Random Forest
Accuracy	86.40%	94.10% ✓
AUC-ROC	0.9340	0.9821 ✓
Precision (Rejected)	0.76	0.93 ✓
Recall (Rejected)	0.72	0.84 ✓
F1-Score (Rejected)	0.74	0.88 ✓
Precision (Approved)	0.90	0.94 ✓
Recall (Approved)	0.92	0.98 ✓
False Positives	75	43 ✓
False Negatives	61	16 ✓



Lower Error Rates

43% reduction in false positives and 74% reduction in false negatives



Better Discrimination

4.8% improvement in AUC-ROC indicates superior ranking ability



Stronger on Minority Class

84% recall on rejected loans vs 72% —critical for risk management

Feature Importance (Random Forest)

Random Forest provides inherent feature importance rankings based on how much each feature contributes to prediction decisions across all decision trees in the ensemble.

Rank	Feature	Importance
1	person_income	38.15%
2	previous_loan_defaults_on_file	22.71%
3	loan_int_rate	12.13%
4	loan_percent_income	10.06%
5	loan_amnt	3.91%
6	person_age	3.02%
7	cb_person_cred_hist_length	2.87%
8	person_emp_length	2.56%

Top 2 Features = 60% Power

Income and prior default history together account for over 60% of the model's predictive capability

Financial Features Dominate

Interest rate and loan-to-income ratio also show strong predictive value

Demographics Minimal

Gender, education, and home ownership contribute very little compared to financial indicators

Key Insight: The model relies primarily on objective financial indicators rather than demographic characteristics, suggesting fair and economically grounded decision-making.

Key Findings

01

Random Forest Superiority

7.7 percentage points higher accuracy and AUC-ROC of 0.98 indicates near-excellent discrimination between approved and rejected loans

02

Exceptional AUC-ROC Performance

0.98 AUC means the model is extremely capable of ranking applicants by probability of approval, making it reliable for risk scoring

03

Financial Features Drive Predictions

Income, default history, interest rate, and loan-to-income ratio dominate. Demographics contribute minimally compared to financial indicators

04

Robust to Class Imbalance

Despite 71/29 class ratio, Random Forest achieves 84% recall on minority class (Rejected) versus Logistic Regression's 72%

05

Logistic Regression Still Viable

Achieves solid results (86.4% accuracy, AUC 0.93) and offers greater interpretability—suitable if explainability prioritized over performance

94.1%

Accuracy

Random Forest performance

0.982

AUC-ROC

Discrimination ability

60%

Top 2 Features

Share of prediction power

84%

Rejected Recall

Minority class capture

Conclusion & Recommendation

Random Forest Classifier

The recommended model for loan approval prediction

Why Random Forest?

The Random Forest model achieves **94.1% accuracy** and an **AUC-ROC of 0.9821**, making it highly reliable for loan approval prediction. It drastically reduces both false positives (risky loan approvals) and false negatives (missed valid applicants) compared to Logistic Regression.



High Accuracy

94.1% overall correctness with strong performance on both classes



Reduced Risk

43 false positives means fewer risky loan approvals



Maximized Opportunity

16 false negatives means nearly all valid applicants approved

Next Steps to Improve Further



Hyperparameter Tuning

Use GridSearchCV to optimize n_estimators, max_depth, and min_samples_split parameters



Address Class Imbalance

Apply SMOTE oversampling or class_weight='balanced' to improve minority class recall



Feature Scaling

Apply StandardScaler to numerical features—required for proper Logistic Regression convergence



Cross-Validation

Implement k-fold CV for more reliable performance estimates



Model Persistence

Export trained model using joblib for deployment in Streamlit application

Report Summary

This analysis evaluated two machine learning models for loan approval prediction using a dataset of 5,000 applications. The Random Forest classifier demonstrated superior performance with 94.1% accuracy and 0.982 AUC-ROC, significantly outperforming Logistic Regression across all metrics.

Key findings reveal that financial features—particularly income and prior default history—drive predictions, while demographic characteristics contribute minimally. The model shows robust performance despite moderate class imbalance, achieving 84% recall on rejected loans while maintaining 94% precision on approved loans.

Recommendation: Deploy Random Forest model with additional hyperparameter tuning and cross-validation for production use in automated loan approval systems.

Report Details

Source: loan_prediction.ipynb

Dataset: 5,000 records

Models: 2 algorithms

Features: 12 variables

Test Set: 1,000 samples

Random Forest achieves 94.1% accuracy with 0.982 AUC-ROC

7.7 percentage points higher than Logistic Regression

43 false positives and 16 false negatives on test set

Top 2 features account for 60% of prediction power